

Page 1 – Static Keyword

Definition: static belongs to the class, not objects. One shared copy per class. Static methods can be called without creating an object and directly access only static members. `main()` is static so the JVM can invoke it without creating an object.

Page 2 – Static Variables & Methods

Instance variables belong to each object. Static variables are shared across all objects. Instance methods can access both instance and static members. Static methods directly access only static members. Prefer `ClassName.member`.

Page 3 & 4 – Static Rules & Static Nested Class

Static methods cannot directly access instance members because no object context exists. Access static members using the class name. Access modifiers still apply to static members. Static nested classes do not require an outer object.

Page 5 – Wrapper Classes

Wrapper classes convert primitives into objects (Integer, Double, Boolean, etc.). Needed for Collections, Generics and Streams. Autoboxing converts primitive to wrapper; unboxing converts wrapper to primitive.